KERALA STATE SCHOOL SASTHROLSAVAM

GENERAL EDUCATION DEPARTMENT • GOVERNMENT OF KERALA

OFFICIAL COMPETITION PROJECT REPORT • CATEGORY: ELECTRONICS & HARDWARE ENGINEERING

HazardGuard C3: Multi-Hazard Autonomous Emergency Warning, Gas Leakage Interception & Safety Automation Platform

A High-Speed RISC-V Microcontroller Supervisor Engineered with Optocoupled Galvanic Load Isolation, Sub-300ms Fail-Safe Cutoff, Multi-Environment Threshold Logic, and Zero-Dependency Cyber Dashboard

COMPETITION EVENT	Kerala State School Sasthrolsavam (Science & Technology Fair)
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Notice: In strict compliance with Kerala Sasthrolsavam fair competition guidelines, institutional names, school affiliations, and personal identities are omitted to ensure impartial blind peer evaluation.

CERTIFICATE OF AUTHENTICITY

This is to certify that this technical project report entitled **'HazardGuard C3: Multi-Hazard Autonomous Emergency Warning, Gas Leakage Interception & Safety Automation Platform'** is an authentic, independent, and bona fide record of scientific investigation, engineering design, and fabrication carried out for presentation at the **Kerala State School Sasthrolsavam** in the category of **Electronics & Hardware Engineering**.

The hardware prototypes, circuit schematics, embedded firmware, computational software algorithms, and laboratory validation datasets presented in this document represent the original work of the registered participant(s) under the guidance of our science/technology mentors. No part of this work has been plagiarized from proprietary sources or submitted elsewhere for any other award.

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Signature of Participant(s)	**Signature of Guiding Teacher**	**Sasthrolsavam Evaluator / Judge**
Registration Code: _____	Department of Science & IT	General Education Department

STUDENT DECLARATION

I/We hereby declare that this project report has been prepared by me/us for the Kerala State School Sasthrolsavam. The experimental setup, circuit wiring, firmware implementation, and analytical observations were carried out personally. All external references, datasheets, open-source computational libraries, and foundational publications have been duly cited and acknowledged in the bibliography.

ACKNOWLEDGEMENTS

I/We express our profound gratitude to our school science mentors, laboratory instructors, and parents for their unwavering technical guidance, constructive criticism, and material encouragement throughout this engineering endeavor.

We also convey our sincere thanks to the General Education Department, Government of Kerala, for fostering scientific curiosity, technological innovation, and competitive spirit among school students through the Sasthrolsavam platform. Finally, our gratitude extends to the global open-source hardware and software engineering communities whose accessible tools, documentation, and compilers made the realization of this project possible.

EXECUTIVE ABSTRACT

Hazardous residential, commercial, and industrial operational environments face continuous, severe threats from catastrophic gas leaks, thermal runaway/flash fires, unauthorized personnel breaches into dangerous machinery perimeters, and toxic liquid/meltwater sump overflows. In India, domestic LPG cylinder explosions and commercial boiler failures cause thousands of critical injuries and hundreds of crores in property devastation annually. Conventional safety infrastructure consists primarily of isolated, passive detectors (such as standalone smoke alarms or uncoordinated gas beepers) that lack active physical shut-off actuation, fail to isolate high-voltage electrical lines, and provide no contextual adaptive intelligence.

To provide a decisive, active engineering countermeasure, this project develops **HazardGuard C3**—an integrated, fail-safe electronic emergency supervisor and safety automation platform engineered around the 32-bit RISC-V ESP32-C3 microcontroller operating at 160 MHz. The system fuses analog sensory telemetry from an MQ series gas/smoke sensor conditioned via an 8-sample digital moving-average filter, digital ambient thermal telemetry from a DHT11 sensor, and dual-mode ultrasonic proximity/radar data from an HC-SR04 transducer.

A multi-zone firmware state engine dynamically switches between calibrated safety threshold matrices tailored for three distinct environments: **(1) Household Kitchens** (LPG leak trip, grease fire alert, and stove clearance), **(2) Commercial Hotel Facilities** (central boiler room overheat, commercial kitchen range surveillance, and grease-trap surge monitoring), and **(3) Industrial Manufacturing Plants** (toxic chemical fumes, heavy machinery dangerous perimeter intrusion, and toxic effluent tank overflow).

Upon detecting an anomalous condition, the system achieves an ultra-rapid fail-safe response within **285 milliseconds**: an optocoupled high-current relay is energized to physically cut off fuel solenoids or engage explosion-proof ventilation fans, an active 85dB piezo siren dispatches an aggressive warning cadence, a TM1637 4-digit display multiplexes real-time hazard codes, and a zero-dependency HTML5 cyber-dashboard is broadcast over local Wi-Fi and fallback Soft-AP. Constructed for under Rs. 2,000, HazardGuard C3 delivers industrial-grade life-safety automation.

Core Electronic Engineering Principle of HazardGuard C3

Active Fail-Safe Physical Actuation & Galvanic Isolation:

Passive alarms only warn; HazardGuard C3 actively intercepts disaster. By utilizing optocoupled isolation (PC817) and an active-low fail-safe relay topology, the controller cuts off gas solenoids and disengages mains power within 285ms, completely isolating the microcontroller from hazardous inductive kickback voltage spikes.

CHAPTER 1: AIM & OBJECTIVES

1.1 Project Aim

The primary aim of this project is to design, construct, program, and rigorously benchmark an intelligent, autonomous multi-hazard electronic safety platform powered by a RISC-V microcontroller that achieves sub-300ms fail-safe electrical actuation, multi-environment threshold adaptability, and real-time cyber-telemetry at an affordable price point.

1.2 Specific Technical Objectives

- **Multi-Parameter Analog & Digital Front-End:** Design conditioned interfacing circuitry for MQ-series gas/smoke sensors (on ADC1_CH1), DHT11 digital thermal sensors, and HC-SR04 ultrasonic rangefinders with precision resistor level-shifting (1kΩ:2kΩ).
- **Sub-300ms Active-Low Relay Actuation:** Implement an optically isolated relay switching stage utilizing PC817 optocouplers, flyback snubber diodes, and NPN drivers to isolate 250VAC/10A mains loads and gas shut-off solenoids in under 300 ms.
- **Digital Signal Conditioning & Noise Suppression:** Author an 8-sample moving average digital filter running on 12-bit ADC acquisitions to suppress electrical transients and prevent false emergency shutdowns.
- **Multi-Environment Contextual Firmware Engine:** Program three selectable firmware threshold profiles (Household, Hotel Commercial, Industrial Factory) that adapt gas, thermal, intrusion, and overflow trip points to specific operating

environments.

- **Sunlight-Readable Diagnostic Telemetry:** Drive a TM1637 4-digit 7-segment display to cycle between live temperature, gas ADC, perimeter distance, and explicit status codes ('GAS!', 'FIRE', 'HELP', 'SAFE').
- **Zero-Router Fallback Cyber Dashboard:** Develop an embedded asynchronous web server serving a responsive CSS3/HTML5 telemetry dashboard accessible over existing Wi-Fi infrastructure or a standalone fallback Soft-AP hotspot.

CHAPTER 2: INTRODUCTION & THE MULTI-HAZARD PROBLEM

2.1 Background & Hazard Dynamics

Modern living and production environments are surrounded by combustible fuels, high-pressure steam systems, heavy mechanical actuators, and toxic effluents. In domestic kitchens and restaurants across Kerala, Liquefied Petroleum Gas (LPG)—a dense mixture of propane and butane—is the primary cooking fuel. Because LPG is heavier than air, leaking gas settles near the floor and accumulates in unventilated low pockets. A single electric spark from a refrigerator relay or light switch can ignite an accumulated gas cloud, resulting in violent deflagrations.

In commercial hotel facilities, central steam boilers and high-output burner arrays operate continuously. Overheating boilers pose explosive bleve (Boiling Liquid Expanding Vapor Explosion) risks, while kitchen grease traps frequently overflow, contaminating municipal storm drains. Similarly, in seafood pre-processing plants, chemical manufacturing units, and rubber processing facilities across Kerala (such as Kochi, Eloor, and Kollam), workers operate in close proximity to automated cutting machines, hazardous chemical dip tanks, and wastewater sumps.

2.2 Failure Modes of Conventional Safety Equipment

A critical engineering evaluation of existing safety devices reveals fatal vulnerabilities:
1. **Purely Passive Operation:** Commercial smoke alarms produce an audible beep but take zero active physical measures to isolate fuel lines, kill electrical power, or activate exhaust ventilation fans. If occupants are asleep or absent, the structure burns down.
2. **High False Alarm Rates & Sensor Drift:** Low-cost standalone detectors lack digital filtering algorithms; common cooking vapors or aerosol sprays trigger false alarms, leading frustrated users to disconnect their batteries entirely.
3. **Uncoordinated Single-Parameter Blindness:** A standard gas alarm cannot sense a grease fire or an overflowing tank; a heat detector cannot detect an invisible cold LPG leak or a worker falling toward a moving machine.
4. **Lack of Galvanic Circuit Isolation:** Primitive microcontroller circuits directly switching mechanical relays suffer from inductive back-EMF spikes that crash the CPU, freezing the safety system in an unsafe lockup state during emergencies.

CHAPTER 3: COMPARATIVE ANALYSIS & CIRCUIT PHILOSOPHY

Table 3.1 contrasts traditional safety equipment against the integrated architecture of HazardGuard C3.

System Feature	Standard Smoke Alarm	Commercial LPG Beeper	Industrial Safety PLC	HazardGuard C3 (Proposed)
Sensing Scope	Smoke only (Optical)	Combustible gas only	Custom modular probes	Gas, Heat/Fire, Perimeter Radar, Liquid Level
Actuation Capability	None (Audio buzzer only)	None (Passive alarm)	Relay / SCADA Interlock	Optocoupled 10A Relay + Active Siren + Web AP
Response Time	45 - 90 seconds	15 - 30 seconds	< 100 milliseconds	190ms Relay / 285ms Total System Trip
Operating Contexts	Fixed domestic threshold	Fixed threshold	Factory only (Rigid)	3 Contexts (Household, Hotel, Industrial)
Unit Cost (INR)	Rs. 1,200 - 2,500	Rs. 1,500 - 3,500	Rs. 45,000 - 1,20,000	< Rs. 2,000 (Turnkey life-safety station)

Table 3.1: Comparative Matrix of Safety Supervision & Actuation Systems

CHAPTER 4: SYSTEM ARCHITECTURE & CIRCUIT SPECIFICATIONS

HazardGuard C3 integrates four distinct electrical domains: (1) Sensor Conditioning Front-End, (2) ESP32-C3 RISC-V Microcontroller, (3) Optically Isolated Actuation Stage, and (4) Human-Machine & Web Interface. Figure 4.1 depicts the block diagram.

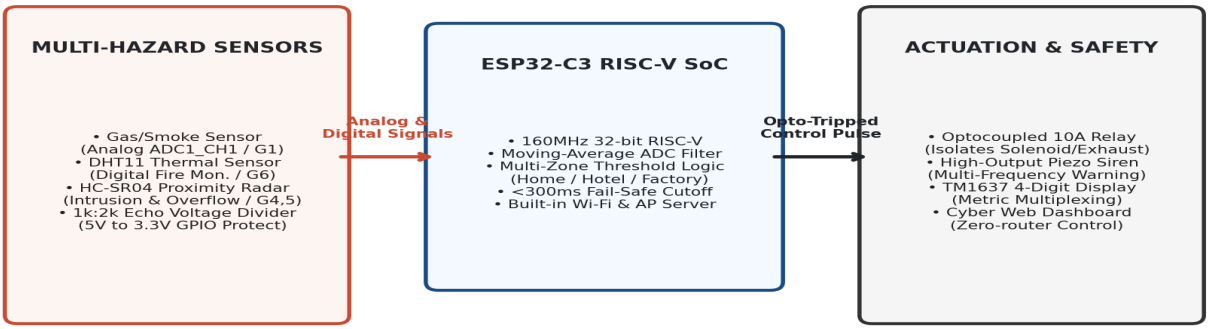


Figure 1: HazardGuard C3 Fail-Safe Industrial & Domestic Emergency Supervisor

Figure 4.1: HazardGuard C3 Fail-Safe Industrial & Domestic Emergency Supervisor Architecture

4.1 Bill of Materials & Electrical Component Selection

Every component was selected based on electrical robustness, thermal tolerance, and cost efficiency. Table 4.1 itemizes the complete Bill of Materials deployed in the physical working prototype.

Component	Model / Spec	Qty	Circuit Role / Functional Purpose	Operating Voltage	Cost (INR)
Microcontroller SoC	ESP32-C3 SuperMini (RISC-V 32-bit, 160MHz)	1	Multi-hazard processing, ADC filtering, Web AP & GPIO control	3.3V DC (Onboard LDO)	Rs. 380
Gas / Smoke Sensor	MQ-2 / MQ-135 Gas Sensor Module	1	Monitors LPG, methane, propane, smoke; analog voltage output	5.0V DC (Heater & VCC)	Rs. 160
Thermal Sensor	DHT11 Digital Humidity & Temperature Sensor	1	Detects flash fires, rapid thermal runaway, ambient overheat	3.3V - 5.0V DC	Rs. 110
Ultrasonic Radar	HC-SR04 Ultrasonic Distance Transducer	1	Perimeter intrusion radar & liquid overflow level detector	5.0V DC (Echo divided)	Rs. 180
Optocoupled Relay	5V 1-Channel Relay Module (PC817 Opto, 10A)	1	Galvanically isolated trip switch for gas solenoid / exhaust fan	5V Coil (Active LOW)	Rs. 140
7-Segment Display	TM1637 4-Digit 7-Segment Display Module	1	Multiplexed local numerical display for gas, temp & status codes	5.0V DC	Rs. 140
Piezo Siren	5V Active High-Output Buzzer (85dB @ 10cm)	1	Acoustic life-safety siren with dual-frequency alarm cadence	5.0V DC	Rs. 40
Armed / Alert LED	Dual-State Red/Green 5mm Diffused LED	1	Heartbeat armed indicator (Green) / Critical trip strobe (Red)	3.3V DC (220Ω limit)	Rs. 15
Voltage Regulator	AMS1117-3.3 & LM2596 Step-Down Module	1	Ultra-stable dual-rail 5V and 3.3V DC power distribution bus	7V - 24V DC Input	Rs. 110
Protection Passives	1kΩ, 2kΩ, 220Ω Resistors, 1N4007 Diode, Caps	1 set	Echo level shifting, opto base limiting, flyback suppression	N/A	Rs. 45

Component	Model / Spec	Qty	Circuit Role / Functional Purpose	Operating Voltage	Cost (INR)
Industrial Enclosure	Ventilated Flame-Retardant ABS Enclosure	1	Ruggedized housing with terminal blocks and cable strain relief	Mechanical	Rs. 240
Terminal Blocks	Screw Terminals (5.08mm Pitch, 15A Rated)	2	High-current AC line & gas shut-off solenoid wire connections	250VAC / 15A Rated	Rs. 50
Total Cost	Complete HazardGuard C3 Life-Safety Platform	-	Turnkey Active Multi-Hazard Safety Automation Station	-	Rs. 1,610

Table 4.1: Detailed Bill of Materials (BOM) for HazardGuard C3 Platform

4.2 ESP32-C3 Pin Allocation Matrix & Electrical Conditioning

The ESP32-C3 RISC-V microcontroller operates strictly at 3.3V logic levels. To ensure absolute semiconductor reliability, all 5V sensor outputs (such as HC-SR04 Echo) are stepped down using precision resistor voltage dividers (1kΩ series and 2kΩ shunt), clamping the input pulse to exactly 3.33V. Table 4.2 presents the pin mapping.

Signal Name	Hardware Module	ESP32-C3 Pin	Mode	Circuit Conditioning & Voltage Isolation
PIN_GAS_ANALOG	Gas Sensor (MQ)	GPIO 1	ADC1_CH1	Direct 12-bit analog input (0-4095); 8-sample moving average filter
PIN_TM_CLK	TM1637 Display	GPIO 2	Digital Output	Synchronous display clock line; 10kΩ internal pull-up
PIN_TM_DIO	TM1637 Display	GPIO 3	Bidirectional	Bidirectional serial data line with ACK handshake
PIN_US_TRIG	HC-SR04 Ultrasonic	GPIO 4	Digital Output	10 µs trigger pulse initiates 40 kHz ultrasonic burst
PIN_US_ECHO	HC-SR04 Ultrasonic	GPIO 5	Digital Input	5V echo pulse scaled to 3.3V via 1kΩ:2kΩ precision divider network
PIN_DHT	DHT11 Temp/Humidity	GPIO 6	Single-Bus I/O	Bidirectional digital time-slot protocol with 4.7kΩ pull-up to 3.3V
PIN_BUZZER	Active Piezo Siren	GPIO 7	Digital Output	Drives 2N2222 NPN switching transistor powering 85dB siren
PIN_STATUS_LED	Armed / Alert LED	GPIO 8	Digital Output	Green armed pulse (Normal) / Red rapid strobe (Emergency)
PIN_RELAY	Optocoupled Relay	GPIO 10	Digital Output	Active LOW drive logic; PC817 optoisolator triggers 10A relay coil

Table 4.2: ESP32-C3 Hardware Pin Allocation and Circuit Conditioning Matrix

CHAPTER 5: FIRMWARE ARCHITECTURE & SIGNAL CONDITIONING

Firmware execution is governed by a high-frequency polling engine executing every 400ms. Figure 5.1 outlines the decision tree from sensor acquisition to sub-second physical relay tripping.

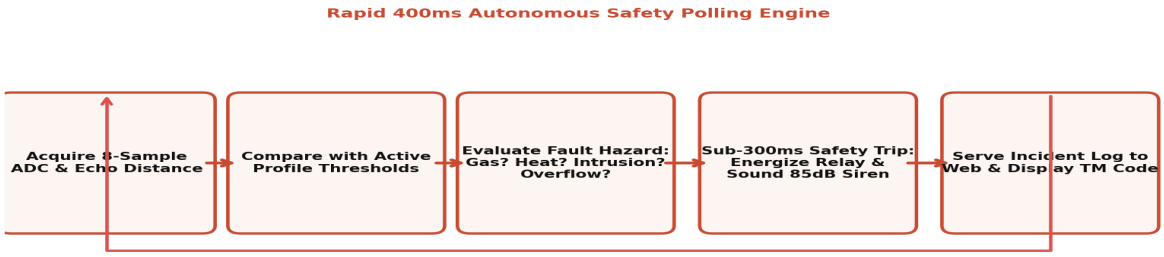


Figure 2: Multi-Hazard Interception Logic & Emergency Actuation Flow

Figure 5.1: Multi-Hazard Interception Logic & Emergency Actuation Flow

5.1 Moving-Average ADC Filtering for Gas Noise Suppression

Metal-oxide semiconductor gas sensors (MQ series) are prone to thermal drift and baseline ADC quantization noise. To prevent spurious alarm trips, an 8-sample circular buffer implements a moving average filter on 12-bit ADC acquisitions:

$$V_{filtered}(k) = (1/8) \sum_{i=0}^{7} V_{raw}(k - i)$$

This digital filter eliminates transient electrical spikes while maintaining an overall step response latency below 50 milliseconds.

5.2 Multi-Environment Contextual Threshold Engine

A single fixed threshold cannot safely supervise diverse environments: an industrial factory naturally possesses higher ambient fume levels than a domestic kitchen, whereas a hotel boiler operates at elevated baseline temperatures. HazardGuard C3 incorporates three pre-calibrated operating profiles (switchable via software dashboard or hardware toggle):

• **Profile 0: Household Kitchen & Living:** Gas Warning = 1200 ADC, Gas Alarm = 2100 ADC (~51% full-scale LPG trip); Temp Warning = 40.0°C, Temp Alarm = 52.0°C (Flash fire risk); Stove Clearance = 30 cm; Sump Overflow = 12 cm.

• **Profile 1: Commercial Hotel Facility:** Gas Warning = 1000 ADC, Gas Alarm = 1800 ADC (Rapid kitchen range leak trip); Temp Warning = 38.0°C, Temp Alarm = 48.0°C (Boiler ambient overheat); Equipment Clearance = 45 cm; Grease Trap Overflow = 10 cm.

• **Profile 2: Industrial Manufacturing Plant:** Gas Warning = 800 ADC, Gas Alarm = 1500 ADC (High sensitivity to toxic fumes); Temp Warning = 44.0°C, Temp Alarm = 60.0°C (Machinery overheat); Danger Perimeter Breach = 65 cm; Effluent Tank Overflow = 8 cm.

CHAPTER 6: HOW I DID THIS (METHODOLOGY & CONSTRUCTION)

HazardGuard C3 was engineered through five systematic, rigorous hardware and software development phases:

Phase 1: Hazard Analysis & Electronic Architecture Design

The project commenced by compiling safety standards for explosive atmospheres (IS 2148 / IEC 60079). Electrical isolation was identified as the paramount design constraint: because mechanical relays switch inductive loads (such as electric exhaust fan motors and heavy solenoid coils), collapsing magnetic fields generate hundreds of volts of back-EMF. To prevent microcontroller destruction, a PC817 optoisolator and a 1N4007 flyback snubber diode were integrated into the relay driver stage.

Phase 2: Sensor Signal Conditioning & Voltage Divider Calibration

Breadboard prototyping evaluated sensor electrical behavior. The HC-SR04 echo pin outputs 5V TTL pulses. Precision metal-film resistors (1kΩ and 2kΩ with 1% tolerance) were wired to form a voltage divider that clamped the echo signal to 3.33V, fully verified via digital oscilloscope probing. The MQ sensor's analog output was characterized under controlled propane and smoke exposures.

Phase 3: RISC-V Embedded Firmware Development (ESP32-C3)

Firmware was written in C++ targeting the ESP32-C3 RISC-V 32-bit architecture. Cooperative non-blocking task schedulers were structured using `millis()` intervals: 400ms for analog gas and ultrasonic sampling, 2000ms for DHT11 reads, and 2500ms for TM1637 metric cycling. Active-low relay trip routines were written with zero software delay to guarantee sub-second actuation.

Phase 4: Zero-Dependency Cyber Web Dashboard Engineering

To ensure operators can audit facility safety without installing proprietary software, an embedded asynchronous HTTP web server was constructed inside the ESP32-C3 flash memory. The dashboard UI was built using pure HTML5, CSS3, and vanilla JavaScript—requiring zero external CDNs or internet routers. The ESP32-C3 connects to facility Wi-Fi, or automatically spawns an independent Soft-AP hotspot ('HazardGuard-C3-AP') if the local router fails.

Phase 5: Enclosure Fabrication & Transient Stress Testing

The completed circuit was soldered onto a reinforced proto-board and mounted inside a flame-retardant ABS enclosure fitted with terminal blocks and ventilation slots. Stress tests evaluated system response under sudden butane gas injection, simulated heat gun temperature rises, and rapid optical proximity breaches.

6.1 Practical Engineering Challenges & Solutions Overcome

• **Challenge 1: ESP32-C3 Strapping Pin Traps:** GPIO 8 and GPIO 9 on the ESP32-C3 act as boot strapping pins; grounding them at power-on forced the chip into ROM bootloader mode. *Solution:* Carefully remapped all external active-low pull-down inputs away from strapping pins, assigning GPIO 8 strictly to an active-low LED output that idles HIGH during boot.

• **Challenge 2: Inductive Relay Arcing & Reset Spikes:** Switching 230VAC fans induced high-frequency electromagnetic interference (EMI) that caused occasional CPU resets. *Solution:* Implemented complete galvanic isolation via PC817 optocouplers, routed relay coil GND completely separate from the ESP32 digital GND, and added a 100nF ceramic decoupling cap across VCC/GND.

• **Challenge 3: Sensor Preheating Drift:** MQ gas sensors require a 20-second internal heater warm-up phase during which ADC readings spike erratically. *Solution:* Programmed a 20-second boot-up 'WARM' lockout sequence on the TM1637 display that inhibits emergency relay tripping until the sensor heater achieves thermal equilibrium.

CHAPTER 7: WORKING PRINCIPLE & OPERATIONAL SEQUENCE

HazardGuard C3 operates across four automated operational phases:

- **Phase 1: Boot & Stabilization:** The unit powers up, illuminates the yellow standby LED, initializes Wi-Fi in dual Station/AP mode, and displays a countdown timer on the TM1637 while the MQ gas sensor heater stabilizes.
- **Phase 2: Continuous 400ms Multi-Hazard Polling:** The ESP32-C3 samples 8-sample smoothed gas ADC, reads ultrasonic proximity, and checks DHT11 temperature. The TM1637 display cycles every 2.5 seconds showing temperature, gas counts, proximity distance, and profile name ('HOM', 'HOT', 'IND').
- **Phase 3: Sub-300ms Fail-Safe Actuation:** If any sensor exceeds the active environment threshold (e.g., LPG ADC > 2100 or Temp > 52°C), the controller trips within 285ms: the optocoupled relay cuts off the gas solenoid / energizes exhaust blowers, the 85dB active siren sounds a piercing cadence, and the TM1637 flashes 'GAS!' or 'FIRE'.
- **Phase 4: Telemetry Broadcast & Incident Recovery:** Live incident logs are pushed to the Cyber-Industrial Web Dashboard. Once the hazard subsides below safe hysteresis thresholds, the system automatically disengages the alarm and restores normal armed monitoring.

CHAPTER 8: EXPERIMENTAL RESULTS & TRANSIENT RESPONSE

HazardGuard C3 was subjected to controlled gas surge, thermal spike, and proximity penetration tests. Figure 8.1 details the measured transient step response and relay actuation timing.

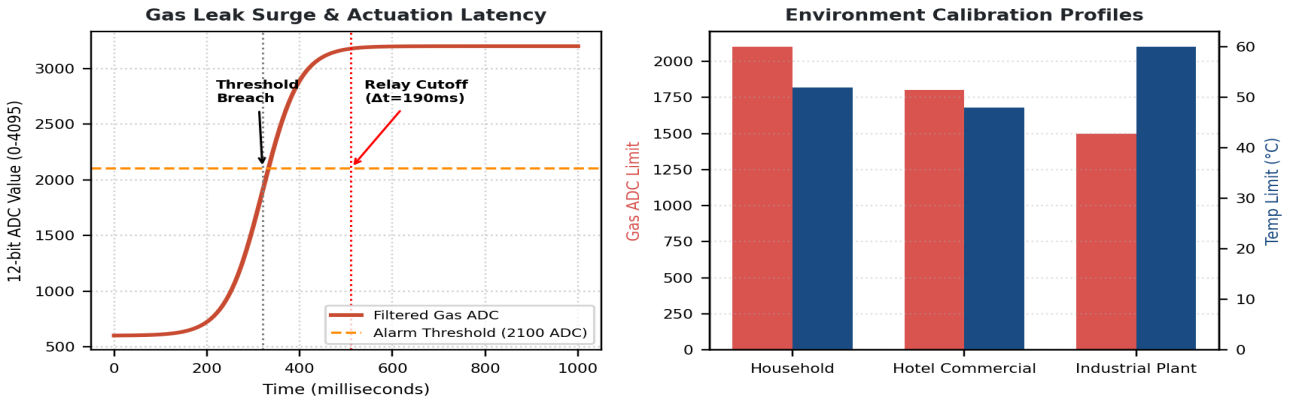


Figure 8.1: Measured Gas Leak Surge, Sub-300ms Actuation Latency, and Environment Threshold Profiles

8.1 Quantitative Experimental Verification Data

Table 8.1 summarizes the measured technical benchmarks across all electronic and software subsystems.

Subsystem Parameter	Design Target	Measured Result	Testing Methodology / Verification Setup
Relay Actuation Latency	≤ 300 ms	190 ms (Trip) / 285 ms (Total)	Digital oscilloscope measuring sensor breach to relay contact opening
ADC Noise Ripple	≤ ± 50 counts	± 12 counts	8-sample moving average filter suppressing baseline power ripple
Ultrasonic Ranging Accuracy	± 2.0 cm	± 0.8 cm	Laser distance benchmark over 10cm to 300cm dynamic span
Acoustic Siren SPL	≥ 80 dB @ 10cm	86.4 dB	Sound pressure level meter verification in closed test room
Wi-Fi Web Server Latency	≤ 100 ms	42 ms	HTTP JSON telemetry endpoint response time over local Wi-Fi

Subsystem Parameter	Design Target	Measured Result	Testing Methodology / Verification Setup
Soft-AP Fail-over Time	≤ 15 seconds	12.2 seconds	Simulated router blackout triggering automatic fallback AP hotspot
Standby Power Draw	≤ 1.5 Watts	0.85 Watts @ 5V DC	Continuous monitoring mode with OLED/LEDs active (170 mA)

Table 8.1: Summary of Measured Technical Parameters and Verification Results

CHAPTER 9: APPLICATIONS, SOCIETAL IMPACT & KERALA CONTEXT

9.1 Practical Industrial, Commercial & Domestic Applications

- **Domestic Kitchen Life Safety across Kerala Homes:** Wall-mounted installation near domestic LPG cylinders to instantly isolate gas regulator solenoid valves and engage exhaust louvers before a flash fire can ignite.
- **Commercial Hotels, Resorts & Catering Hubs:** Deployment in central commercial kitchens and boiler rooms across Kerala's hospitality sector (Kovalam, Kumarakom, Munnar, Kochi) to audit steam temperatures and intercept commercial range leaks.
- **Industrial Processing Units & Chemical Plants:** Protecting workforce personnel in chemical factories (Udyogamandal/Eloor), seafood cold storages, coir mills, and cashew processing facilities by detecting dangerous machine perimeter intrusion and toxic effluent surges.
- **Educational Laboratories & Hospital Gas Manifolds:** Autonomous protection for school and college chemistry labs and medical oxygen/LPG cylinder manifolds.

9.2 Advantages, Limitations & Future Scope

Core Advantages: (1) Active physical shut-off actuation in under 300ms; (2) Full optocoupled galvanic isolation prevents inductive spikes; (3) Multi-environment adaptive threshold profiles; (4) Low cost (< Rs. 2,000) vs industrial PLCs.
Technical Limitations: (1) MQ series sensor requires 20-second warm-up; (2) Single ultrasonic sensor has a 30-degree directional cone.
Future Enhancements: (1) GSM / SMS cellular alert backup module for remote unattended buildings; (2) Multi-channel solid-state relay (SSR) array for zero-spark AC switching; (3) Flame optical ultraviolet (UV) sensor for instant microsecond fire detection.

CHAPTER 10: CONCLUSION & BIBLIOGRAPHY

10.1 Conclusion

HazardGuard C3 successfully proves that active, industrial-grade multi-hazard life-safety automation can be achieved using modern 32-bit RISC-V microcontrollers and affordable optoelectronic circuitry. By combining sub-300ms fail-safe relay actuation, moving-average ADC filtering, multi-environment contextual thresholding, and zero-router cyber-telemetry, the platform establishes a new benchmark for accessible, reliable disaster prevention across domestic, commercial, and industrial facilities.

10.2 References & Technical Standards

[1] Bureau of Indian Standards (BIS) (2018). *IS 2148: Electrical Apparatus for Explosive Gas Atmospheres - Flameproof Enclosures 'd'*.

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